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compact1, compact2, compact3

java.time.chrono

Interface Era

All Superinterfaces:

`TemporalAccessor`, `TemporalAdjuster`

All Known Implementing Classes:

`HijrahEra`, `IsoEra`, `JapaneseEra`, `MinguoEra`, `ThaiBuddhistEra`

```
public interface Era
extends TemporalAccessor, TemporalAdjuster
```

An era of the time-line.

Most calendar systems have a single epoch dividing the time-line into two eras. However, some calendar systems, have multiple eras, such as one for the reign of each leader. In all cases, the era is conceptually the largest division of the time-line. Each chronology defines the Era's that are known Eras and a `Chronology.eras` to get the valid eras.

For example, the Thai Buddhist calendar system divides time into two eras, before and after a single date. By contrast, the Japanese calendar system has one era for the reign of each Emperor.

Instances of `Era` may be compared using the == operator.

Implementation Requirements:

This interface must be implemented with care to ensure other classes operate correctly. All implementations must be singletons - final, immutable and thread-safe. It is recommended to use an enum whenever possible.

Since:

1.8

Method Summary

Abstract Methods | Default Methods

Modifier and Type	Method	Description
default `Temporal`	**`adjustInto`**(`Temporal` temporal)	Adjusts the specified temporal object to have the same era as this object.
default int	**`get`**(`TemporalField` field)	Gets the value of the specified field from this era as an `int`.
default `String`	**`getDisplayName`**(`TextStyle` style, `Locale` locale)	Gets the textual representation of this era.
default long	**`getLong`**(`TemporalField` field)	Gets the value of the specified field from this era as a `long`.
int	**`getValue`**()	Gets the numeric value associated with the era as defined by the chronology.
default boolean	**`isSupported`**(`TemporalField` field)	Checks if the specified field is supported.
default <R> R	**`query`**(`TemporalQuery`<R> query)	Queries this era using the specified query.
default `ValueRange`	**`range`**(`TemporalField` field)	

		Gets the range of valid values for the specified field.

Method Detail

getValue

```
int getValue()
```

Gets the numeric value associated with the era as defined by the chronology. Each chronology defines the predefined Eras and methods to list the Eras of the chronology.

All fields, including eras, have an associated numeric value. The meaning of the numeric value for era is determined by the chronology according to these principles:

- The era in use at the epoch 1970-01-01 (ISO) has the value 1.
- Later eras have sequentially higher values.
- Earlier eras have sequentially lower values, which may be negative.

Returns:

`the numeric era value`

isSupported

```
default boolean isSupported(TemporalField field)
```

Checks if the specified field is supported.

This checks if this era can be queried for the specified field. If false, then calling the `range` and `get` methods will throw an exception.

If the field is a `ChronoField` then the query is implemented here. The `ERA` field returns true. All other `ChronoField` instances will return false.

If the field is not a `ChronoField`, then the result of this method is obtained by invoking `TemporalField.isSupportedBy(TemporalAccessor)` passing `this` as the argument. Whether the field is supported is determined by the field.

Specified by:

`isSupported` in interface `TemporalAccessor`

Parameters:

`field - the field to check, null returns false`

Returns:

true if the field is supported on this era, false if not

range

default ValueRange range(TemporalField field)

Gets the range of valid values for the specified field.

The range object expresses the minimum and maximum valid values for a field. This era is used to enhance the accuracy of the returned range. If it is not possible to return the range, because the field is not supported or for some other reason, an exception is thrown.

If the field is a ChronoField then the query is implemented here. The ERA field returns the range. All other ChronoField instances will throw an UnsupportedTemporalTypeException.

If the field is not a ChronoField, then the result of this method is obtained by invoking TemporalField.rangeRefinedBy(TemporalAccessor) passing this as the argument. Whether the range can be obtained is determined by the field.

The default implementation must return a range for ERA from zero to one, suitable for two era calendar systems such as ISO.

Specified by:

range in interface TemporalAccessor

Parameters:

field - the field to query the range for, not null

Returns:

the range of valid values for the field, not null

Throws:

DateTimeException - if the range for the field cannot be obtained

UnsupportedTemporalTypeException - if the unit is not supported

get

default int get(TemporalField field)

Gets the value of the specified field from this era as an int.

This queries this era for the value of the specified field. The returned value will always be within the valid range of values for the field. If it is not possible to return the value, because the field is not supported or for some other reason, an exception is thrown.

If the field is a ChronoField then the query is implemented here. The ERA field returns the value of the era. All other ChronoField instances will throw an UnsupportedTemporalTypeException.

If the field is not a ChronoField, then the result of this method is obtained by invoking TemporalField.getFrom(TemporalAccessor) passing this as the argument. Whether the value can be obtained, and what the value represents, is determined by the field.

Specified by:

get in interface TemporalAccessor

Parameters:

field - the field to get, not null

Returns:

the value for the field

Throws:

DateTimeException - if a value for the field cannot be obtained or the value is outside the range of valid values for the field

UnsupportedTemporalTypeException - if the field is not supported or the range of values exceeds an int

ArithmeticException - if numeric overflow occurs

getLong

default long getLong(TemporalField field)

Gets the value of the specified field from this era as a long.

This queries this era for the value of the specified field. If it is not possible to return the value, because the field is not supported or for some other reason, an exception is thrown.

If the field is a ChronoField then the query is implemented here. The ERA field returns the value of the era. All other ChronoField instances will throw an UnsupportedTemporalTypeException.

If the field is not a ChronoField, then the result of this method is obtained by invoking TemporalField.getFrom(TemporalAccessor) passing this as the argument. Whether the value can be obtained, and what the value represents, is determined by the field.

Specified by:

getLong in interface TemporalAccessor

Parameters:

field - the field to get, not null

Returns:

the value for the field

Throws:

DateTimeException - if a value for the field cannot be obtained

UnsupportedTemporalTypeException - if the field is not supported

ArithmeticException - if numeric overflow occurs

query

default <R> R query(TemporalQuery<R> query)

Queries this era using the specified query.

This queries this era using the specified query strategy object. The TemporalQuery object defines the logic to be used to obtain the result. Read the documentation of the query to understand what the result of this method will be.

The result of this method is obtained by invoking the TemporalQuery.queryFrom(TemporalAccessor) method on the specified query passing this as the argument.

Specified by:

query in interface TemporalAccessor

Type Parameters:

R - the type of the result

Parameters:

query - the query to invoke, not null

Returns:

the query result, null may be returned (defined by the query)

Throws:

DateTimeException - if unable to query (defined by the query)

ArithmeticException - if numeric overflow occurs (defined by the query)

adjustInto

default Temporal adjustInto(Temporal temporal)

Adjusts the specified temporal object to have the same era as this object.

This returns a temporal object of the same observable type as the input with the era changed to be the same as this.

The adjustment is equivalent to using `Temporal.with(TemporalField, long)` passing `ChronoField.ERA` as the field.

In most cases, it is clearer to reverse the calling pattern by using `Temporal.with(TemporalAdjuster)`:

```
// these two lines are equivalent, but the second approach is recommended
temporal = thisEra.adjustInto(temporal);
temporal = temporal.with(thisEra);
```

This instance is immutable and unaffected by this method call.

Specified by:

`adjustInto` in interface `TemporalAdjuster`

Parameters:

`temporal` - the target object to be adjusted, not null

Returns:

the adjusted object, not null

Throws:

`DateTimeException` - if unable to make the adjustment

`ArithmeticException` - if numeric overflow occurs

getDisplayName

```
default String getDisplayName(TextStyle style,
                              Locale locale)
```

Gets the textual representation of this era.

This returns the textual name used to identify the era, suitable for presentation to the user. The parameters control the style of the returned text and the locale.

If no textual mapping is found then the `numeric value` is returned.

API Note:

This default implementation is suitable for most implementations.

Parameters:

`style` - the style of the text required, not null

`locale` - the locale to use, not null

Returns:

the text value of the era, not null

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